

**EDO STATE COLLEGE OF AGRICULTURE AND NATURAL
RESOURCES, IGUORIAKHI, NIGERIA**

DEPARTMENT OF AGRICULTURAL TECHNOLOGY

COURSE CODE: AGT 215

COURSE TITLE: SOIL FERTILITY AND CROP NUTRITION

CREDIT LOAD: 3 UNITS

INTRODUCTION

This course provides an understanding of the nature and characteristics of soils and plant nutrition. The course is introduced at both theoretical and practical level. Students are prepared to go beyond the normal in-the-classroom learning to field visits. These visits are practically used to further inculcate the classroom teachings into the students on a real-life scenario. The course aims and objectives are provided below:

Course aim and objectives

The course aims at introducing the students to the various aspects of soil, its characterization and its nutrient providing ability to the plants using a theoretical and practical approach to learning. On completion of this course, Students should be able to achieve the following objectives:

- A. Discuss and understand the concept of crop nutrition.
- B. Describe the individual soil characteristics affecting plant growth.
- C. Have an understanding of soil depth, textural and structural soil properties and how they impact fertility.
- D. Understand the influence of soil salinity and acidity on nutrient availability.
- E. Understand soil moisture and why it is important to nutrient availability and uptake.
- F. Discuss soil organic matter and its effects on soil nutrition.
- G. Understand soil organisms and their impact on the fertility of the soil.
- H. Describe the principles and practice of crop nutrition management.

PART 1: CONCEPT OF CROP NUTRITION

Definition of Crop Nutrition

Crop nutrition is the process by which plants obtain and use essential nutrients from soil, air, and water for growth, development, and yield production.

Plants require nutrients for:

- Growth
- Energy production
- Formation of plant tissues
- Reproduction

Sources of Plant Nutrients

Plants obtain nutrients from:

Source	Nutrients
Air	Carbon, Oxygen
Water	Hydrogen
Soil	Nitrogen, Phosphorus, Potassium, etc

Essential Plant Nutrients

There are **17 essential nutrients**, classified into:

Macronutrients (Required in large quantities)

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Calcium (Ca)
- Magnesium (Mg)
- Sulphur (S)

Micronutrients (Required in small quantities)

- Iron (Fe)
- Zinc (Zn)
- Copper (Cu)
- Manganese (Mn)
- Boron (B)
- Molybdenum (Mo)

- Chlorine (Cl)
- Nickel (Ni)

Functions of Essential Nutrients

Nutrient	Function
Nitrogen	Leaf growth
Phosphorus	Root development
Potassium	Disease resistance
Calcium	Cell wall strength
Magnesium	Chlorophyll formation
Iron	Photosynthesis

Deficiency Symptoms

Nutrient	Deficiency symptom
Nitrogen	Yellow leaves
Phosphorus	Poor root growth
Potassium	Weak plants
Iron	Yellowing between veins

INDIVIDUAL SOIL CHARACTERISTICS AFFECTING PLANT GROWTH

Soil characteristics determine fertility and productivity.

Physical Characteristics

Soil Texture

This is the Proportion of sand, silt, and clay in a soil.

Types:

- Sandy soil
- Clay soil
- Loamy soil

Suitable soil for Agricultural Production: Loamy soil

Advantages of loamy soil:

- Good drainage
- Good nutrient retention

Soil Structure

This has to do with the arrangement of soil particles.

For Example;

- Granular
- Blocky
- Platy

Good structure improves:

- Root growth
- Water movement
- Air circulation

Soil Depth

Deep soils support:

- Better root growth
- Higher productivity
 - Shallow soils reduce yield.

Chemical Characteristics

Soil pH

Soil acidity or alkalinity.

Scale: 0–14

Neutral = 7

Most crops grow best at pH 5.5–7.0

Soil Organic Matter

Importance:

- Improves fertility
- Improves water retention
- Improves soil structure

Biological Characteristics

Includes:

- Microorganisms
- Earthworms

Functions:

- Nutrient cycling
- Organic matter decomposition

SOIL DEPTH, TEXTURE, AND STRUCTURE AND THEIR EFFECT ON FERTILITY

Effect of Soil Depth

Deep soil:

- High water storage
- Good root development
- High fertility

Shallow soil:

- Low fertility
- Poor plant growth

Effect of Soil Texture

Texture	Fertility
Sandy	Low fertility
Clay	High fertility
Loamy	Best fertility

Effect of Soil Structure

Good structure improves:

- Root penetration
- Nutrient uptake
- Water availability

PRINCIPLES AND PRACTICES OF CROP NUTRITION MANAGEMENT

Fertility Management Methods

Organic Fertilizers

Organic fertilizers are natural materials derived from plant and animal sources that supply essential nutrients to crops and improve soil fertility and structure.

They release nutrients slowly and improve long-term soil health.

Examples include:

- Compost
- Farmyard manure
- Poultry manure

COMPOST

Definition

Compost is decomposed organic material made from plant and animal wastes through the action of microorganisms under controlled conditions.

It is a dark, crumbly, soil-like material rich in nutrients.

Materials Used to Make Compost

Plant Materials

- Dry leaves
- Grass clippings
- Crop residues
- Kitchen wastes (vegetable peels)

Animal Materials

- Animal dung
- Animal urine

Other Materials

- Wood ash
- Sawdust

Composting Process

Composting occurs in stages:

Stage 1: Collection of materials

Organic wastes are gathered.

Stage 2: Decomposition

Microorganisms break down organic materials.

Stage 3: Heat production

Temperature rises to 50–70°C, killing pathogens.

Stage 4: Maturation

Compost cools and becomes ready for use.

Types of Composting

Heap composting: Materials piled on ground.

Pit composting: Materials placed in pits.

Modern composting: Uses compost bins and controlled aeration.

Nutrient Content of Compost

Compost contains:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Organic matter

Benefits of Compost

- Improves soil fertility
- Adds nutrients.
- Improves soil structure
- Improves aeration.
- Improves water retention
- Promotes microbial activity
- Environmentally friendly

Application Method

Applied by:

- Broadcasting
- Mixing with soil
- Applied before planting

FARMYARD MANURE (FYM)

Definition

Farmyard manure is decomposed mixture of animal dung, urine, bedding materials, and leftover feed.

It is obtained from farm animals such as:

- Cattle
- Sheep
- Goat

Composition of Farmyard Manure

Contains:

- Animal dung
- Urine
- Straw
- Feed residues

Nutrient Content of FYM

Average nutrient content:

- Nitrogen: 0.5–1%
- Phosphorus: 0.2–0.5%
- Potassium: 0.5–1%

Preparation of Farmyard Manure

Steps:

1. Collect dung and urine
2. Store in pit or heap
3. Allow decomposition for 2–4 months

Benefits of Farmyard Manure

- Improves soil fertility
- Improves soil structure
- Improves soil water retention
- Improves soil microbial activity
- Improves crop yield

Methods of Application

- Broadcasting
- Band placement
- Incorporated into soil

Modern Importance of FYM

Used in:

- Organic farming
- Sustainable agriculture
- Climate-smart agriculture

POULTRY MANURE

Definition

Poultry manure is waste obtained from poultry birds such as:

- Chicken
- Turkey
- Duck

It includes:

- Droppings
- Feathers
- Bedding materials

Nutrient Content of Poultry Manure

Poultry manure is richer than other manures.

Contains:

- Nitrogen: 2–4%
- Phosphorus: 2–3%
- Potassium: 1–2%

Characteristics of Poultry Manure

- Fast nutrient release
- High fertility value
- Strong smell
- High nitrogen content

Advantages of Poultry Manure

- Improves crop growth rapidly
- Improves soil fertility
- Improves soil structure
- Improves crop yield

Disadvantages

If used fresh, it can:

- Burn crops
- Cause nutrient imbalance

Therefore, it should be cured before use.

Preparation of Poultry Manure

Steps:

1. Collect manure
2. Allow decomposition for 2–3 weeks
3. Apply to soil

Methods of Application

- Broadcasting
- Mixing with soil
- Applied before planting

Application rate: 5–10 tons per hectare

COMPARISON OF ORGANIC FERTILIZERS

Fertilizer	Nutrient Level	Speed of Release
Compost	Moderate	Slow
Farmyard manure	Moderate	Slow
Poultry manure	High	Fast

IMPORTANCE OF ORGANIC FERTILIZERS IN MODERN AGRICULTURE

They:

- Improve soil health
- Improve soil structure
- Increase crop yield

- Reduce chemical fertilizer use
- Improve environmental sustainability

ROLE IN CLIMATE SMART AGRICULTURE

Organic fertilizers:

- Reduce soil degradation
- Improve carbon storage
- Reduce environmental pollution

Inorganic Fertilizers

Examples:

- Urea
- NPK
- SSP

Advantages:

- Fast nutrient supply

Methods of Fertilizer Application

- Broadcasting
- Band placement
- Side dressing
- Foliar application

Integrated Soil Fertility Management (ISFM)

Combination of organic and inorganic fertilizers.

Benefits:

- Improve productivity
- Improve sustainability

Modern Nutrient Management Techniques: Precision Agriculture

Uses:

- Soil testing
- GPS
- Sensors

Benefits:

- Efficient fertilizer use
- Reduce waste

Climate Smart Fertility Management

Includes:

- Organic fertilizer use
- Crop rotation
- Cover cropping

Soil Testing

Importance:

- Determines soil nutrient status
- Helps fertilizer recommendation

PART 2: Understanding the Influence of Soil Salinity and Acidity on Nutrient Availability

1. Concept of Soil Salinity and Acidity

1.1 Soil Salinity

Soil salinity refers to the accumulation of soluble salts such as sodium (Na^+), calcium (Ca^{2+}), magnesium (Mg^{2+}), chloride (Cl^-), sulphate (SO_4^{2-}), and bicarbonate (HCO_3^-) in the soil solution up to a level that adversely affect plant growth and soil productivity. Soil salinity is measured using the electrical conductivity meter. A soil is considered saline when the **electrical conductivity (EC) of its saturation extract exceeds 4 decisiemens/meter (dS/m) at 25°C**, though many sensitive crops may show symptoms at lower levels. High salinity causes osmotic stress, making it difficult for plants to absorb water and nutrients.

1.2 Soil Acidity

Soil acidity refers to the concentration of hydrogen ions (H^+) and aluminium ions (Al^{3+}) in the soil solution and it is usually expressed as soil pH.

- **Strongly acidic soils** have pH values below 5.5.
- **Moderately acidic soils** range from pH 5.5 - 6.5, while neutral soils are around pH 7.0.

Soil acidity influences chemical reactions, nutrient solubility, and microbial activity in the soil. Acidic soils are common in humid tropical regions due to high rainfall and leaching.

2. Causes of Soil Salinity and Acidity

2.1 Causes of Soil Salinity

1. Natural (Primary) Causes

- Weathering of parent materials that are rich in soluble salts.
- Capillary rise of saline groundwater in arid and semi-arid areas.
- Poor natural drainage leading to salt accumulation at the surface.

2. Human-Induced (Secondary) Causes

- Excessive irrigation, especially with saline water.
- Inadequate drainage in irrigated fields.

- Use of high-sodium fertilizers or effluents.
- Sea water intrusion in coastal soils.

2.2 Causes of Soil Acidity

1. **Leaching of Basic Cations such as Ca^{2+} , Mg^{2+} , K^+ , Na^+ :** Heavy rainfall leaches away base cations, leaving acidic ions (H^+ and Al^{3+}) dominant in the soil.
2. **Use of Acid-Forming Fertilizers:** Ammonium-based fertilizers (e.g., ammonium sulphate, urea) release H^+ ions during nitrification, thereby lowering pH and causing soil acidity.
3. **Organic Matter Decomposition:** Microbial breakdown of organic residues produces organic acids that acidify the soil.
4. **Parent Material:** Soils derived from acidic rocks (like granite or quartzite) tend to be naturally acidic when weathering of these parent materials occur.
5. **Acid Rain and Industrial Pollution:** Deposition of sulphuric and nitric acids from polluted air contributes to soil acidification.

3. Effects of Soil Salinity and Acidity on Nutrient Availability

3.1 Effects of Soil Salinity

- **Reduced Nutrient Uptake:** High osmotic potential limits water and nutrient absorption by roots.
- **Ion Imbalance and Toxicity:** Excess Na^+ and Cl^- compete with essential nutrients like K^+ , Ca^{2+} , and NO_3^- , causing deficiencies.
- **Reduced Solubility of Some Nutrients:** Phosphorus may precipitate as calcium phosphate in saline soils, reducing its availability.
- **Micronutrient Deficiencies:** Fe, Zn, and Mn availability decreases under saline conditions, particularly in calcareous soils.

3.2 Effects of Soil Acidity

- **Reduced Availability of Macronutrients:** Phosphorus reacts with Fe and Al in acidic soils to form insoluble phosphates. Calcium, magnesium, and molybdenum availability also decrease at low pH.

- **Increased Availability of Toxic Elements:** Soluble aluminium (Al^{3+}), manganese (Mn^{2+}), and iron (Fe^{2+}) may reach toxic levels in strongly acidic soils, damaging roots.
- **Reduced Microbial Activity:** Acidic conditions suppress beneficial soil microorganisms, particularly nitrifiers, thereby reducing N mineralization and availability.
- **Nutrient Imbalance:** Excess H^+ and Al^{3+} ions displace base cations from exchange sites, causing nutrient imbalance and poor plant growth.

4. Impact of Corrective Treatments on Soil Salinity and Acidity

4.1 Corrective Treatments for Salinity

1. **Leaching:** Applying high-quality (non-saline) irrigation water to flush out excess salts below the root zone.
2. **Improved Drainage:** Installation of drainage systems to prevent salt buildup.
3. **Application of Soil Amendments:** Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) is commonly used to replace exchangeable Na^+ with Ca^{2+} , improving soil structure and permeability.
4. **Use of Salt-Tolerant Crops:** Cultivating crops such as barley, cotton, and date palm in saline soils.
5. **Mulching and Organic Matter:** Organic mulches improve infiltration, reduce evaporation, and enhance microbial activity.

Effect on Nutrient Availability:

- Gypsum improves Ca^{2+} and S availability and enhances root uptake of other nutrients.
- Leaching reduces toxic ions (Na^+ , Cl^-), restoring nutrient balance.
- Improved soil structure enhances aeration and root access to nutrients.

4.2 Corrective Treatments for Acidity

1. **Liming:** Application of agricultural lime (CaCO_3) or dolomite ($\text{CaMg}(\text{CO}_3)_2$) neutralizes soil acidity and supplies Ca^{2+} and Mg^{2+} .
2. **Use of Acid-Tolerant Crops:** Crops such as cassava, sweet potato, and tea perform relatively well in acidic soils.
3. **Balanced Fertilization:** Avoiding excessive ammonium-based fertilizers and applying basic fertilizers like calcium nitrate.

4. **Organic Matter Addition:** Organic materials buffer pH, increase cation exchange capacity, and chelate toxic metals.

Effect on Nutrient Availability:

- Liming increases the availability of P, Mo, Ca, and Mg while reducing Al and Mn toxicity.
- Microbial activity improves, enhancing N mineralization and nutrient cycling.
- The overall nutrient-use efficiency of crops increases, leading to better growth and yields.

5. Note

- **Soil salinity** and **acidity** are major constraints to nutrient availability and crop productivity.
- Salinity causes osmotic stress, ion toxicity, and nutrient imbalance, while acidity leads to nutrient fixation, toxicity, and microbial inhibition.
- Corrective measures such as leaching, liming, gypsum application, and organic amendments help restore soil fertility by improving nutrient solubility and uptake.
- Sustainable soil management practices that prevent salinization and acidification are essential for maintaining long-term soil productivity and environmental quality.

Understanding Soil Moisture and Its Importance

1. Definition of Soil Moisture

Soil moisture refers to the amount of water present in the soil pores or spaces between soil particles. It is the portion of water that the soil holds after rainfall, irrigation, or condensation. This water serves as a critical link between the soil, plants, and atmosphere—supporting plant growth, microbial activity, and nutrient transport.

Soil moisture exists in different forms, ranging from freely draining water to tightly bound films around soil particles. Its quantity and availability depend on soil texture, structure, organic matter content, and climatic conditions.

2. Importance of Soil Moisture to Plant Nutrition

Soil moisture is one of the most important factors influencing plant growth and nutrient uptake. Its major roles include:

1. **Medium for Nutrient Transport:** Nutrients in the soil dissolve in water, forming a soil solution. Roots absorb these nutrients in dissolved form, meaning that without adequate soil moisture, nutrient uptake is restricted.
2. **Facilitates Chemical Reactions:** Many soil chemical and biological reactions such as mineralization, nitrification, and decomposition require water to occur efficiently.
3. **Maintains Soil Microbial Activity:** Microorganisms responsible for nutrient cycling (e.g., nitrogen-fixing bacteria, decomposers) thrive in moist soils. Excessively dry or waterlogged conditions reduce microbial activity.
4. **Aids Photosynthesis and Transpiration:** Water absorbed by roots is used in photosynthesis and also regulates temperature through transpiration.
5. **Supports Root Growth:** Adequate soil moisture softens the soil, allowing easy root penetration and exploration of a larger soil volume for nutrient uptake.

3. Different Classes of Soil Moisture

Soil moisture can be divided into different classes based on how strongly water is held in the soil and its availability to plants:

Class	Description	Availability to Plants
Gravitational Water	Water that drains through the soil under the influence of gravity, occupying large pores (macropores) immediately after rainfall or irrigation.	Unavailable , as it drains too quickly for roots to absorb.
Capillary Water	Water held in small soil pores (micropores) by surface tension, forming a film around soil particles.	Available , as plants can absorb it easily.
Hygroscopic Water	Thin film of water tightly bound to soil particle surfaces due to adhesion forces.	Unavailable , as it cannot be absorbed by roots.

Chemically Bound Water	Water held within the crystal structure of minerals.	Permanently unavailable.
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4. Available and Unavailable Forms of Soil Moisture

4.1 Available Forms

The **available water** for plant use lies between:

- **Field Capacity (FC):** The amount of water retained in the soil after gravitational drainage has stopped (about 1–3 days after heavy rain).
- **Permanent Wilting Point (PWP):** The moisture level at which plants can no longer extract water and begin to wilt permanently.

Available Water Capacity (AWC) = Field Capacity – Permanent Wilting Point.

This range represents the **usable moisture reserve** that sustains plants between irrigation or rainfall events.

4.2 Unavailable Forms

- **Gravitational Water:** Quickly lost after rain or irrigation and thus not available to plants.
- **Hygroscopic Water:** Held too tightly by soil particles (especially in clay soils) for plant roots to extract.
- **Chemically Bound Water:** Locked in mineral structures and cannot be used by plants.

5. Importance of Soil Moisture on Nutrient Availability to Crops (With Simple Experiment)

5.1 Concept

Soil moisture directly influences nutrient mobility, root absorption, and microbial processes that release nutrients from organic and mineral sources. When soil is too dry, diffusion and mass flow of nutrients slow down. When soil is waterlogged, air supply is cut off, reducing root respiration and microbial activity.

5.2 Simple Classroom or Field Experiment

Title: Effect of Soil Moisture on Nutrient Availability and Plant Growth

Objective:

To demonstrate how different levels of soil moisture affect nutrient uptake and plant growth.

Materials Needed:

- Three pots (A, B, C) filled with the same type of loamy soil
- Uniform seedlings (e.g., maize or beans)
- Watering can
- Measuring cylinder
- Labels for each pot

Experimental Setup:

Pot	Treatment	Moisture Condition
A	Dry soil	No watering after planting
B	Optimum moisture	Watered to field capacity daily
C	Waterlogged soil	Kept saturated with water

Procedure:

1. Plant equal-sized seedlings in all three pots.
2. Maintain moisture treatments as described above for 2–3 weeks.
3. Observe growth performance, leaf color, and vigor.

Expected Results:

- **Pot A (Dry):** Plants show stunted growth, yellow leaves, and poor nutrient uptake due to limited dissolution and movement of nutrients.
- **Pot B (Optimum):** Plants exhibit healthy growth, dark green leaves, and vigorous development, indicating adequate nutrient availability.
- **Pot C (Waterlogged):** Plants show yellowing or wilting due to oxygen deficiency, root decay, and reduced uptake of nutrients like nitrogen and potassium.

Conclusion:

Optimum soil moisture enhances nutrient solubility, root activity, and microbial mineralization—thereby improving nutrient availability and plant growth. Both drought and waterlogging negatively affect nutrient use efficiency.

6. Note

- Soil moisture is the water held within soil pores and is essential for nutrient transport, root function, and microbial activity.
- The **available water** for plants lies between **field capacity** and **permanent wilting point**.
- Both excess and deficiency of soil moisture reduce nutrient availability and crop productivity.
- Proper soil moisture management ensures efficient use of fertilizers, healthy root growth, and improved yields.

Understanding Soil Organic Matter and Its Effects on Soil Properties

1. Description of Soil Organic Matter

Soil Organic Matter (SOM) refers to all organic materials found in the soil, including living organisms (microbes, plant roots), freshly added residues (crop or animal waste), actively decomposing materials, and highly decomposed stable compounds such as humus. SOM plays a key role in soil formation, nutrient cycling, and overall soil productivity. It influences both the chemical and physical characteristics of soil, improving its structure and enhancing the environment for plant growth.

2. Sources of Soil Organic Matter

Organic matter in soil originates from both living and dead organisms. Major sources include:

- **Plant residues** such as leaves, stems, and roots left on or within the soil after harvest
- **Animal manures** including dung, urine, and bedding materials
- **Green manure crops**, incorporated while still fresh
- **Compost materials** derived from decomposed organic wastes

- **Soil microorganisms** and soil fauna (fungi, bacteria, earthworms)
- **Humus**, which forms after prolonged decomposition

These materials undergo microbial breakdown, contributing nutrients to the soil and forming the stable organic fraction.

3. Factors Affecting Quantity of Organic Matter in Soil

The amount of SOM present depends strongly on environmental and management factors:

- **Climate:** Cooler, wetter climates promote accumulation. Warm, dry climates accelerate decomposition and reduce SOM.
- **Soil texture:** Clay-rich soils protect organic molecules from breakdown and retain more SOM than sandy soils.
- **Vegetation type:** Deep-rooting perennial grasses contribute more SOM compared to shallow rooting crops.
- **Land use and management:** Continuous cropping, burning crop residues, and excessive tillage lower SOM. Conservation agriculture builds SOM.
- **Drainage and aeration:** Poorly aerated, waterlogged soils slow decomposition, increasing SOM levels.
- **Organic inputs:** Addition of manures, compost, and crop residues increases SOM.

4. Types of Organic Matter and Their Contribution to Soil Fertility

a. Green Manure

Green manure refers to fresh plant material grown and incorporated directly into the soil.

Contributions:

- Rapid nutrient release, especially nitrogen from legumes.
- Improves soil structure and reduces erosion.
- Enhances microbial activity and organic carbon supply.
- Increases moisture retention in sandy soils.

b. Farmyard Manure (FYM)

FYM consists of animal dung, urine, and bedding materials collected from livestock housing.

Contributions:

- Adds both macro- and micronutrients in slow-release forms
- Improves soil aggregation and tilth
- Enhances soil water-holding capacity
- Stimulates microorganism populations and nutrient cycling

c. Compost

Compost is well-decomposed organic waste material produced under controlled conditions.

Contributions:

- Supplies stable nutrients and organic carbon
- Enhances cation exchange capacity (CEC)
- Improves soil pH buffering capacity
- Supports long-term fertility and disease suppression

5. Importance of Stable Humus to Soil Fertility and Plant Growth

Humus is the dark, highly decomposed fraction of SOM that resists further breakdown. Its roles include:

- **Improved nutrient retention** due to high CEC
- **Enhanced soil structure** through formation of stable aggregates
- **Improved water infiltration and storage**
- **Buffering of soil pH**, protecting plants from acidity or alkalinity extremes
- **Immobilization of toxic elements**, creating a safer root environment
- **Long-term carbon storage** which sustains soil productivity over time

Humus acts like a soil “bank” that stores nutrients and releases them slowly to plants.

6. Effect of Levels of Organic Matter on Soil Health

Soil health is strongly tied to organic matter levels:

- **High SOM levels:** Promote good soil structure, aeration, and biodiversity. They support strong root systems, efficient nutrient cycling, and resilience against drought or erosion.
- **Low SOM levels:** Lead to compacted, nutrient-poor soils with low microbial activity. Crops become more susceptible to stress and yield declines.

Maintaining SOM is considered one of the most important indicators of sustainable soil management and long-term agricultural productivity.

Note

Soil Organic Matter is a foundation of fertile, healthy soil. Its presence improves physical structure, enhances chemical fertility, and supports biological life. Through appropriate management, particularly by returning organic inputs like green manure, FYM, and compost, farmers can build resilient soils that nourish crops and sustain productivity far into the future.

Understanding Soil Organisms and Their Impact on Soil Fertility

1. Introduction

Soil is a living system that contains a wide range of organisms, from microscopic bacteria to larger animals like earthworms and rodents. These organisms interact with each other and with soil components, driving nutrient cycling, improving soil structure, and promoting healthy plant growth. Their activities form the biological foundation of soil fertility and long-term productivity.

2. Soil Macrofauna

Soil macrofauna are organisms large enough to be seen with the naked eye, generally greater than 2 mm in size. Common examples include:

a. Earthworms

- Often called “ecosystem engineers”
- Its burrowing activities improves soil aeration and drainage
- Can cause fragmenting of plant residues thereby enhancing decomposition

- Casts are rich in available nutrients like nitrogen and phosphorus

b. Rodents (e.g., squirrels, rats)

- Its burrowing activities improves aeration and water infiltration
- Can aid mixing of organic matter and soil layers
- However, may damage crops and expose soil to erosion if overabundant

c. Soil-dwelling mammals (e.g., moles, rabbits)

- Improve soil structure through tunneling
- Their waste contributes organic matter and nutrients
- Can disturb plant roots if populations are uncontrolled

d. Snakes

- Predators that help regulate populations of rodents and insects
- Maintain ecological balance indirectly supporting soil ecosystem health

e. Termites and Crickets

- Break down tough plant residues such as wood and dry grass
- Termite mounds enhance soil nutrient heterogeneity and microstructure
- Promote microbial activity through organic decomposition

Overall role of macrofauna: They create biopores, enhance soil aggregation, accelerate organic matter breakdown, and improve nutrient availability for plant uptake.

3. Soil Microflora

Soil microflora represents microscopic plant-like and microbial inhabitants of soil. They include bacteria, algae, fungi, and actinomycetes. These organisms drive decomposition, nutrient transformation, and soil biochemical processes critical to plant growth.

a. Bacteria

- Most abundant soil organisms
- Conduct key nutrient transformations such as nitrogen fixation, nitrification, and denitrification
- Decompose simple organic compounds
- Increase nutrient availability, especially nitrogen for crops

b. Algae

- Photosynthetic organisms in soil surface layers
- Produce organic matter and oxygen in the soil environment
- Some cyanobacteria fix atmospheric nitrogen, enriching infertile soils (e.g., paddy fields)

c. Fungi

- Decompose complex organic materials such as cellulose and lignin
- Form symbiotic associations like mycorrhizae that enhance water and phosphorus uptake
- Improve soil aggregation through hyphal networks

d. Actinomycetes

- Filamentous bacteria that thrive in well-drained soils
- Responsible for final stages of decomposition and humus formation
- Produce antibiotics that suppress harmful pathogens
- Contribute to soil structure and long-term fertility

4. How Microflora Contribute to Soil Fertility

Microflora activities support:

- Continuous release of nutrients from organic matter
- Formation of stable humus improving Cation Exchange Capacity (CEC)
- Biological nitrogen inputs (especially from N-fixing microbes)

- Disease suppression in the rhizosphere
- Breakdown of pesticides and pollutants, improving soil quality

5. Overall Influence of Soil Organisms on Soil Productivity

Soil organisms are essential drivers of soil health. Their collective activities:

- Improve soil physical structure through aggregation and pore creation
- Accelerate nutrient cycling for plant uptake
- Enhance soil water storage and infiltration
- Support root growth through improved aeration and reduced compaction
- Promote biological diversity and ecosystem balance

High levels of biological activity are a strong indicator of fertile and productive soils. When soil organisms are negatively affected by poor land management, fertility declines and crop productivity suffers.

Note

Soil organisms—from tiny microbes to large burrowing animals—play interconnected roles that sustain soil fertility. Maintaining a healthy biological community through organic inputs, reduced tillage, and conservation practices ensures long-term soil productivity and sustainable agriculture.